Digestive System - Gastrointestinal Tract

DIGESTIVE SYSTEM

The digestive system consists of the digestive tract and its associated glands. The digestive tract is a long tube that begins at the mouth and ends at the anus. The tube starts in the oral cavity and continues as the esophagus, stomach, small intestine, large intestine, rectum and anal canal. The large glands lie outside the digestive tract but have ducts which open into the tube. They are the salivary glands, pancreas, liver and gallbladder.

The entire tube is lined by a **mucosa**, which consists of a lubricated epithelium with an underlying thin layer of vascular connective tissue (**lamina propria**) and a thin layer of smooth muscle, the **muscularis mucosae**. Deep to the mucosa, is the **submucosa** made of dense irregular connective tissue, blood vessels and a nerve plexus (**submucosal plexus** or Meissner's). Surrounding the submucosa are bands of smooth muscle, the **muscularis externa**. These bands usually run in two directions, inner circular and outer longitudinal. Between these bands is another nerve plexus, the **myenteric plexus** (Auerbach's). Outside of the muscularis externa is a connective tissue layer- either a **serosa** or an **adventitia**. A serosa has a thin, simple squamous epithelium – a mesothelium – while an adventitia has no epithelial lining.

Laboratory Experience

The purpose of this laboratory exercise is to identify the cells and structures associated with digestive organs. You will identify the different cell types present, their cytological structure, the supporting connective tissues of the organs that comprise the digestive tract.
Use the slide images listed below along with your manual, your text and your atlas to identify the structures described.

Organs presented in this interactive laboratory session:

Identify and check-off each of the following:

☐ **Tongue**	☐ **Appendix**
☐ **Esophagus**	☐ **Anal canal**
☐ **Stomach**	☐ **Liver**
☐ **Small intestine**	☐ **Salivary glands**
☐ **Colon**	☐ **Pancreas**
	☐ **Gall bladder**

Lingual Papillae: Slide #2-3 (H&E)

The bulk of the **tongue** consists of skeletal muscle whose fibers cross each other in three planes. They are grouped in bundles separated by connective tissue. The dorsal surface of the tongue is covered by a stratified squamous epithelium; separated into an anterior and posterior region by a V-shaped depression, the sulcus terminalis. Papillae cover the dorsal surface of the tongue anterior to the sulcus terminalis.

1) **Filiform Papillae** - numerous elongated cones of epithelium and connective tissue that give the tongue its rough surface. The filiform papillae are sharp, often partly keratinized conical projections; these are the most numerous and lack taste buds. x45

2) **Fungiform Papillae** - mushroom-shaped projections with narrow stalks and wide upper parts. They occur singly and are scattered throughout the filiform papillae and may contain taste buds on their dorsal surface.

3) **Foliate Papillae** – parallel low ridges separated by deep mucosal clefts. They occur on the lateral edge of the tongue and contain many taste buds in the clefts facing neighboring papillae.

4) **Circumvallate Papillae** – large, round projections surrounded by moat-like structures. There are 7-12 of them arranged in a V-shaped configuration separating the anterior from the posterior region of the tongue. Their lateral surfaces contain numerous **taste buds. Serous lingual glands** (of Von Ebner) empty into the moat-like structures to maintain a continuous flow of fluid over the taste buds. Additional taste buds are found scattered throughout the oral cavity; in the tongue, epiglottis, pharynx and palate.

Identify: Filiform papillae, fungiform papillae, foliate papillae, circumvallate papillae, taste buds

Overview of the Gastrointestinal tract: Slide # 4 Diagram outlining basic composition of the walls of the gastrointestinal tract.

Esophagus: Slide #5-7 (Trichrome stain)

Mucosa:
Mucosa is composed of epithelium, lamina propria and muscularis mucosa (also called muscularis interna).

- Epithelium: Several layers of cells. Most basal layer of cells is columnar and as they ascend are cuboidal and the most apical layers of cells are squamous. The basal cell layer is attached to basement membrane by means of hemidesmosomes. The most apical cells retain their nuclei. Therefore, the epithelium is defined as stratified squamous non-keratinized epithelium, which is characteristic of moist epithelium, which is also adapted for wear and tear.
- Lamina propria: characterized by the presence of connective tissue with few defensive cells. Mucus glands, especially towards the lower end of esophagus close to its termination are called the esophageal cardiac glands. These provide slimy mucus that enables the food to slide off into the stomach.

- Muscularis mucosae is very thick distally and consist mainly of longitudinal smooth muscle fibers.

Submucosa:

- Forms longitudinal ridges when the tube is contracted.
- Is made of dense irregular connective tissue.
- It contains scattered mucous glands throughout the length of the esophagus that serve to lubricate the epithelium. These are called the esophageal glands proper. Esophagus is one of the two regions in the tubular tract that contains submucosal glands, the other being duodenum.
- Also characterized by the presence of large blood vessels, nerves (Meissner's plexus) and lymphatics.

Muscularis externa:

- The esophageal wall is composed of two thick muscle layers, arranged as inner circular and outer longitudinal layers. Auerbach's plexus can often be observed between the two layers.
- In the upper third of the esophagus both layers are composed of skeletal muscle.
- The middle third contains skeletal and smooth muscle fibers and, in the lower third, mainly smooth muscles.
- Circular bundles of smooth muscle provide a physiological sphincter-like function at the distal end of the esophagus.

Adventitia / serosa:

- Above the diaphragm, there is an adventitia consisting loose connective tissue while, below the diaphragm, there is a **serosa (peritoneal lining)**.

Identify: Mucosa, epithelium, lamina propia, muscularis mucosae, submucosa, esophageal glands proper, Meissner's plexus (if present), muscularis externa, inner circular layer, outer longitudinal layer, Auerbach's plexus, adventitia/serosa

Esophago-gastric junction: Slide #8 (H&E)

Lower end of the esophagus passes through the diaphragm at the level of T 10 vertebra and opens through the cardiac orifice into the cardiac portion of the stomach.

The following changes occur in the four layers of the wall:

- Mucosa:

  The esophageal epithelium which is non-keratinized, stratified squamous changes abruptly at the esophago-gastric junction to a simple columnar epithelium (primarily mucous secreting cells) with simple tubular or branched tubular glands. In the distal esophagus and upper cardiac regions are collections of mucus-secreting glands called **esophageal cardiac glands** (because they resemble the cardiac glands of the stomach). Muscularis interna develops an additional layer forming inner circular and outer longitudinal arrangement of smooth muscles.

- Submucosa:

  Mucus glands start to disappear.

- Muscularis externa:

  The circular muscle layer acts as a physiological sphincter and prevents regurgitation of food once it enters the stomach. Additional innermost oblique layer of muscle starts developing.

- Serosa:

  Remains the same.

Stomach

Slide 09: Fundus, H&E, medium magnification, inset shows rugae at fundus, H&E, low magnification,
Slide 10: Fundic gastric glands, Alcian blue/PAS x 320 with illustration
Slide 11: Fundus, H&E, high magnification
Slide 12: Cardiac glands, H&E x240; Pyloric glands, H&E x120

Three regions of the stomach are recognized histologically; **cardiac, body / fundus and pylorus**. Each of these regions has a simple columnar epithelium, but the type and quantity of cells that make up the epithelium is quite different.

Throughout the stomach, the **mucosa** contains a simple columnar epithelium that is invaginated into the lamina propria forming **gastric pits** which further split into many tubular **gastric glands**. The lamina propria is loose connective tissue with immune cells. The well-developed muscularis mucosa is found at the base of the glands and separates the mucosa from the submucosa. The gastric epithelium turns over rapidly with the surface cells being replaced every 4 - 7 days while the deep cells are replaced much more slowly.

The **submucosa** contains dense irregular connective tissue with blood vessels, lymphocytes, macrophages and mast cells. It produces ridges or folds called **rugae** that are smoothed out when the stomach is distended. Large blood vessels, Meissner's nerve plexus and lymphatics are also seen in the submucosa.

Three thick layers of smooth muscle (outer longitudinal, middle circular and inner oblique) form the **muscularis externa**, although distinct layers may be difficult to discern. The middle circular layer thickens to participate in the formation of the pyloric sphincter. Between the circular and longitudinal layers of muscles, Auerbach's plexus may be observed.

The **serosa** is covered by a mesothelium that is continuous with the gastric mesenteries.

Fundic Stomach

Of the three regions of the stomach, the fundus / body is the largest region of the stomach.

Mucosa:

The surface epithelium is a simple columnar epithelium that produces shallow **gastric pits (**by invaginating into the lamina propria). From the gastric pits, the epithelium divides into long, simple tubular **gastric glands** that contain a neck and a base.

1. Mucus cells:
   The surface epithelium and gastric pits are made of **surface mucous cells** that secrete a thick, viscous mucous to protect the epithelial lining. At the base of the gastric pits and in the neck of the gastric glands are found **mucous neck cells** that produce a thin, watery mucous for transport of the other gastric gland secretions.

2. Stem cells:
   The isthmus of the gastric gland also contains **stem cells** that divide and produce all the other epithelial cell types.

3. Parietal/Oxyntic cells:
   Within the neck and base of the gastric glands are found the **parietal** (oxyntic) **cells**. These cells are large, pyramidal, and acidophilic with central, round nucleus. The electron microscope reveals intracellular canaliculi lined with abundant microvilli. Parietal cells make and secrete separately **hydrogen and chloride ions** that produce hydrochloric acid (with less than pH 2).

   Parietal cells also produce **gastric intrinsic factor**, a glycoprotein which binds vitamin B12 in the lumen of the stomach for absorption in the ileum. Acid secretion by parietal cells is stimulated by cholinergic nerve endings, by histamine and by gastrin.

4. Chief/ Zymogenic cells:
   Primarily found in the base of the gastric glands are **chief** (zymogenic) **cells**. These basophilic cuboidal-to-low columnar cells contain apical, membrane-bound secretion granules containing pepsinogen, lipase and other digestive enzymes in an inactive form. These are secreted and then activated by the low pH that is found in the lumen of the gland.

5. APUD / Enteroendocrine Cells:
   Also found in the base of the gastric glands (as well as throughout the rest of the tubular digestive system) are **enteroendocrine** (argentaffin, APUD) **cells**. These cells have an affinity for silver stains and are hormone secreting cells. They secrete their products into the lamina propria and not into the lumen of the digestive tract. Therefore, their apex is towards the lamina propria with their base on the luminal surface. Hormones produced by enteroendocrine cells of the stomach include 5-hydroxytryptamine (serotonin), gastrin and glucagon. Gastrin is produced by cells in the pyloric portion of the stomach; it stimulates HCl and pepsin synthesis.

The submucosa, muscularis externa and serosa of the fundic stomach are the same as described under the general features of the stomach.
In addition, when the stomach is empty, the submucosa is thrown into folds called rugae.

At the pyloric antrum, the middle (circular) layer of smooth muscle of the muscularis externa is thickened to form the **pyloric sphincter**.

Identify: Mucosa, epithelium, gastric pits, gastric glands, mucous cells, chief cells, parietal cells, enteroendocrine cells, lamina propia, muscularis mucosae, submucosa, Meissner's plexus, muscularis externa, Auerbach's plexus, serosa

Gastroduodenal junction

Slide 13: Gastroduodenal junction, H&E x40, x120, x640

The pylorus of the stomach transitions to the duodenum of the small intestine. The following changes can be observed during this transition:

- Mucosa:
  The epithelium remains simple columnar. The mucus secreting cells are replaced by columnar absorptive cells. Invagination of this epithelium into the lamina propria forms intestinal glands and villi can be observed.

- Submucosa:
  Mucus-secreting glands (of Brunner) are a distinct feature of the submucosa of the duodenum.

- Muscularis externa:
  Innermost oblique layer seen in the stomach disappears. Muscularis externa of the small intestine includes the inner circular and outer longitudinal layers; Auerbach's plexus may be observed.

- Serosa:
  In some areas devoid of peritoneum, this outermost layer is adventitia.

Small Intestine

Slide 14: Duodenum, H&E x120
Slide 15: Jejunum, H&E x22, plica circularis, H&E x60
Slide 16: Small intestine: Villus x500, Goblet cells, Brush border, PAS+ hematoxylin high magnification
Slide 17: Jejunum, Paneth cells, H&E x40, inset x680
Slide 18: Ileum, longitudinal H&E x40 and cross-section x20

The small intestine is a narrow tube about 4 meters long, divided into three regions: the duodenum, jejunum and ileum. The final digestion of chime and the absorption of small molecules take place in the small intestine.

Mucosa:
The hallmark of the small intestine is the presence of **intestinal villi** which are finger-like outgrowths of the mucous membrane into the lumen. The plica circularis, the villi and the microvilli on the absorptive cells all increase the surface area for absorption. These surface area specializations decrease distally in the small intestine, so that the ileum has a smaller surface area than the jejunum or duodenum.

The epithelium of the villi in the small intestine contains two major cell types; the columnar absorptive cells and the goblet cells.

1) Columnar absorptive cells:
These have regular, closely-packed microvilli on their apical surfaces and tight junctional complexes around their apical borders. The lateral membranes from adjacent columnar cells form complex interdigitations. Each columnar cell secretes its own apical surface coat of glycoprotein enzymes which, along with pancreatic enzymes, complete the chemical breakdown of food. Through the apical membrane, amino acids, fatty acids and monosaccharides, along with water, minerals and electrolytes enter the cell by both passive and active transport.

Monoglycerides and fatty acids are esterified to triglycerides in the smooth endoplasmic reticulum, coated with specialized protein and released as chylomicrons into the extracellular space where they are transported by lacteals. These cells also synthesize a glycoprotein which attaches to secretory IgA from plasma cells and this complex is released into the lumen.

2) Goblet cells:
The goblet cells are unicellular, mucous secreting cells that protect and lubricate the epithelial lining. They become more numerous distally in the small intestine. Therefore, there are more goblet cells in the epithelium of the ileum than in the jejunum or duodenum.

Around the base of each villus, the epithelium descends into several short tubular glands into the lamina propria, called **intestinal glands** (Crypts of Lieberkühn).

The simple columnar epithelium of the villus is continuous with that lining the glands. The epithelium of the glands contains absorptive cells and goblet cells as well as other epithelial cells; stem cells, Paneth cells and enteroendocrine cells.

3) Stem cells:
Stem cells are small, undifferentiated columnar cells that divide and differentiate into the other cell types.

4) Paneth cells:
Paneth cells are large pyramidal cells found at the base of the glands with bright, eosinophilic granules. These granules contain lysozyme which is secreted into the lumen of the small intestine and acts as a bacteriocide.

5) Enteroendocrine / APUD cells:
These cells in the small intestine are like those found in the stomach. They secrete specific hormones into the lamina propria for local control of digestion. Hormones produced by the enteroendocrine cells of the small intestine include secretin, cholecystokinin and motilin.

The **lamina propria** of the mucosa of the small intestine penetrates the core of the villi and is composed of blood vessels, lymphatics, immune cells, fibroblasts and smooth muscle cells. The blood vessels consist of fenestrated capillaries. A central lacteal in the villus absorbs chylomicrons into the lymphatic circulation.

Muscularis mucosae: Smooth muscle fibers course up into each villus and insert on the subepithelial connective tissue. These cause the villus to contract several times per minute.

Plicae circularis:

The **mucosa**, along with submucosa of the small intestine has a series of permanent folds, **plica circularis/Valves of Kerckring**. These modifications increase the surface area of absorption.

Submucosa:

- The **submucosa** of the small intestine is composed of dense irregular connective tissue as well as immune cells, blood vessels and nerve cell bodies / fibers. The nerve cell bodies contain postganglionic parasympathetic cell bodies in the **submucosal plexus** (Meissner's).

- In the duodenum, **submucosal glands** (of Brunner) are connected to the lumen by ducts that drain into the base of the crypts. The alkaline mucus secretion from the glands neutralizes the acidic chyme entering the duodenum.

- In the ileum, submucosal lymph nodules are common and aggregate to form **Peyer's patches**.

Muscularis Externa:

The **muscularis externa** of the small intestine contains an inner circular and outer longitudinal arrangement of smooth muscle. Between the layers of smooth muscle is found a neural plexus called myenteric **plexus** (Auerbach's) with postganglionic parasympathetic cell bodies.

Serosa /adventitia:
The jejunum and the ileum of the small intestine are intraperitoneal and, therefore, are surrounded by a **serosa**.
The first and fourth portions of the duodenum are covered by a serosa while the second and third portions are retroperitoneal. Therefore, these regions have **serosa** on the anterior surface with an **adventitia** on the posterior surface towards posterior abdominal wall.

Distinguishing Features of the Regions of the Small Intestine

- Duodenum:
  The duodenum is distinguished by the appearance of villi and the presence of the **submucosal glands (of Brunner)**.

- Jejunum:
  The jejunum has prominent **plicae circulares** and long villi, a greater number of goblet cells and no submucosal glands.

- Ileum:
  The ileum is distinguished by large aggregates of lymphatic nodules known as **Peyer's patches**. Plicae circulares become reduced in size and frequency; villi become shorter; even more goblet cells.

Identify: Mucosa, epithelium, microvilli, villi, intestinal glands, enterocytes, goblet cells, Paneth cells, lamina propia, muscularis mucosae, submucosa, plicae circularis, Meissner's plexus, muscularis externa, Auerbach's plexus, serosa

Large Intestine, colon

Slide 19: Colon, H&E x30 and x140

The large intestine extends from the ileum to the anus and consists of the cecum, appendix, ascending colon, transverse colon, descending colon, sigmoid colon, rectum and anal canal. It is about 1.5 meters long and twice the width of the small intestine. The large intestine normally absorbs large amounts of water and small amounts of sugars and vitamins (e.g., vitamin K). Histologically, the large intestine is similar throughout its length.

Mucosa:
The mucosa of the large intestine is flat and undistinguished. There are no villi present and only straight tubular **intestinal glands** (Crypts of Lieberkühn) are found.

- Epithelium:
  The epithelium of the large intestine is composed mainly of **goblet cells** with interspersed **columnar absorptive cells** and a few **enteroendocrine cells** found in the base of the glands. The epithelium is continuously renewed by mitosis of undifferentiated cells in the lower half of the glands.
- The lamina propria:
  Is rich in immune cells and lymph nodules. The nodules frequently invade the submucosa.

Submucosa:
Contains many lymph nodules which are especially abundant in the appendix. Rest of the components like lymphatics, nerve plexus and blood vessels are same as the other parts of the tubular GI.

Muscularis externa:
Have inner circular and outer longitudinal smooth muscle fibers. The outer longitudinal layer is gathered into three thick longitudinal bands called **taeniae coli**. The portions of the wall without the longitudinal layer are weakened and project outwards as **haustra**. These features are very typical for the large intestine.

Serosa:
The large intestine has a **serosa** on the cecum, appendix, transverse colon and sigmoid colon. Since the ascending colon and descending colon are retroperitoneal, they have a **serosa** on the anterior surface and an **adventitia** on the posterior surface. The serosa of the large intestine contains small pendulous protuberances of adipose tissue called **appendices epiploicae**.

Appendix
Slide # 20: Appendix, H&E x25, x80, inset x200

The **appendix** is a blind, finger-like extension of the cecum that often has its lumen obliterated by abundant lymphoid follicles and sometimes with fecal matter.

These enlarged lymphoid nodules create a stellate lumen appearance in cross section profiles. Its general structure is like that of the large intestine, but it contains fewer, shorter intestinal glands and has no taeniae coli.

The taenia coli converge at the base of the appendix, which is one of the features used during surgery to locate the appendix.

Identify: Mucosa, epithelium, intestinal glands, enterocytes, goblet cells, lamina propia, muscularis mucosae, submucosa, Meissner's plexus, muscularis externa, Auerbach's plexus, serosa

Anal Canal
Slide 21 & 22: Anal canal, H&E x10 and x200 with illustration.

The **anal canal** is the distal portion of the gastrointestinal tract. In the terminal rectum and the initial part of the anal canal, there are a few simple tubular glands with a simple columnar epithelial lining composed almost entirely of goblet cells. The columnar epithelium of the colorectal zone (CRZ) transitions to a stratified cuboidal or columnar epithelium of the anal transition zone (ATZ) which transitions to the stratified squamous nonkeratinized epithelium of the squamous zone (SQZ) lining the rest of the anal canal. This is relatively short and at the anal orifice, the epithelium changes again to epidermis (stratified squamous keratinized) and deep to it are the large circumanal glands.

- The proximal half of the anal canal contains several longitudinal folds, produced by an infolding of the mucous membrane around the submucosa, known as the rectal columns (of Morgagni).

- They are separated from each other by furrow-like rectal sinuses which end distally in small valve-like folds called the anal valves.

- The pectinate line from gross anatomy refers to the area between the anal transition zone and the squamous zone.

- The **submucosa** contains a large plexus of veins which form hemorrhoids when excessively dilated.

- The internal anal sphincter corresponds to the inner circular layer of the muscularis externa. The external anal sphincter is skeletal muscle.

Identify: CRZ, ATZ, SQZ, anal glands, internal anal sphincter, external anal sphincter

Glands associated with the gastrointestinal tract

Liver

Slide #24 Illustration showing arrangement of tissue and blood supply

Slide #25 Pig liver, Mallory-Azan x65; Human liver, H&E x65

Slide #26 Liver lobules diagram

Slide #27 Portal triad, Mallory trichrome

Slide #28 Liver, H&E x500, inset x800; Toluidine blue x900

Slide #29 Hepatic sinusoids, TEM x6600; Perisinusoidal space, TEM x18000

Slide #30 Bile canaliculi, H&E x1240; Canals of Hering, H&E x800

There are two organizations that will become apparent:

Most of the tissue is comprised of eosinophilic **epithelial hepatocytes** organized as plates**.**

Scattered among the epithelia clusters are islands of connective tissue that contain a branch of the **portal vein, hepatic artery, and bile duct.** These islands are the **portal triads.** The triads range in size from large triads (about 34 hepatic lobules in diameter) to small triads that mark the periphery of the hepatic lobules.

Some of the liver lobules will be cut in cross-section. In these lobules, the liver plates radiate from a **central vein** whose intima resembles a sinusoid (there is no observable media or adventitia). Frequently, **sinusoids** can be seen to be continuous with the central veins.

At the periphery of a lobule you will likely find 3 to 6 portal triads, each enveloped in a CT sleeve.

In some sections, the endothelial **cells** of the sinusoid are quite evident. Hepatic sinusoids are lined by a discontinuous epithelium with large fenestrae and a discontinuous basal lamina. **Stellate sinusoidal macrophages (Kupffer cells)** may often be observed in the sinusoids.

Note that a space exists between the endothelial cells and hepatocytes. This is the **perisinusoidal space of Disse. Hepatic stellate cells (Ito cells)** may be found in the perisinusoidal space and function to store vitamin A; however, in some pathological cases Ito cells can differentiate into myofibroblasts and secrete collagen.

The largest vessel in each triad is the portal vein. The bile duct is distinguished by its simple cuboidal epithelium. Identify a branch of the hepatic artery and see if you find a lymphatic in there as well.

The biliary tree is a 3-dimensional system of channels. **Bile canaliculi** are the smallest branches of the biliary tree formed by the apposed grooves of adjacent hepatocytes. Hepatocytes secrete bile into the canaliculi. Near the portal canal, bile canaliculi become the **canals of Hering.**

Canals of Hering are lined by hepatocytes and cholangiocytes and are thought to be a reservoir of liver progenitor cells. Bile flows from canals of Hering to **intrahepatic bile ductules** (lined by cholangiocytes) to **interlocular bile ducts** to the **right & left hepatic ducts** to the **common hepatic duct.**

Identify and check-off each of the following:

( ) Portal triad	**( ) Portal vein**	**( ) Hepatic artery**	**( ) Lymphatics**
( ) Sinusoids ( ) Bile duct	**( ) Hepatocytes**	**( ) Central vein**	
( ) Space of Disse			

Gallbladder

Slide #31 Gallbladder, H&E x175 , inset H&E x325 and illustration of absorptive cells

The gallbladder functions to store and concentrate bile. The mucosa of the gall bladder is thrown up into folds. In situ the mucosa will exhibit this appearance when the gall bladder is empty. The folds disappear as the lumen becomes distended with bile.

The lining is a simple columnar epithelium with numerous, but short microvilli. Beneath the epithelium is the lamina propria, which consists of a cellular, loose connective tissue containing large numbers of lymphocytes and plasma cells. Under this layer is the muscularis externa, which when stimulated will compress the lumen and force the bile out into the bile duct.

External to the muscularis externa is the adventitia for attachment to the liver; or the serosa for the unattached surface.

Identify and check-off each of the following:

( ) Simple columnar epithelium	**( ) Lamina propria**	**( ) Muscularis externa**	**( ) Serosa**

Pancreas

Slide #32 Pancreas, H&E x25

Slide #33 Pancreas, H&E x860 and pancreatic acinus, TEM x 5800

Connective tissue from the capsule penetrates the gland and subdivides it into lobules.

At higher magnification examine the serous acini that comprise the lobule. The acinar cells exhibit the character of protein secretors: rounded euchromatic nuclei, basal basophilia, and apical accumulations of eosinophilic secretory granules. There are no mucus units in the pancreas.

In the acini identify centroacinar cells. They are characterized by their large pale staining nuclei found in the central region of the acinus.

Intralobular ducts are found among the acini. They are narrow and lined by a low cuboidal epithelium; they possess a wide lumen.

The larger intralobular ducts are frequently enmeshed in collagen. Interlobular ducts and large blood vessels are readily visible within thick interlobular CT sheaths.

Note that the pancreas lacks striated ducts and myoepithelial cells – with your knowledge analyze why.

Identify and check-off each of the following:

( ) CT capsule ( ) CT septum ( ) Intercalated duct ( ) intralobular duct
( ) Interlobular duct ( ) Serous acini ( ) Centroacinar cells

Organization of the major salivary glands

Slide # 34 Illustration showing arrangement of secretory units and ducts of slivary gland

Sublingual Gland

Slide #35- Sublingual, Mallory trichrome

Glandular Epithelium

The sublingual gland is a branched tubuloacinar gland comprised of mostly mucus and some serous cells. There are no serous acinar units. The serous cells when present are attached to the ends of the tubular mucus glands as demilunes, crescent-shaped appendages.

The connective tissue forms a capsule around the gland and it provides thin septa that subdivide the secretory units into lobules.

The interlobular septa are best observed at low magnification, though the CT is quite apparent around the larger interlobular ducts.

The secretory units are mostly columnar mucus secretors, as is evidenced by their cytology: flocculent, pale stained cytoplasm and basally compressed, condensed nuclei. The narrow, centrally placed lumens of the secretory units are usually difficult to observe. Widely scattered crescent-shaped clusters of basophilic serous cells form “caps” upon the mucus glands.

Duct Epithelium

Within the lobules are numerous epithelial ducts. The smaller intercalated ducts are simple cuboidal, larger intralobular or striated ducts become simple columnar. At the basal margin of the striated ducts the cytoplasm is striped. These basal striations reflect mitochondrial-filled folds in the basal plasmalemma, increased surface area for active ion transport.

The largest ducts are found in the interlobular CT. These interlobular ducts are generally lined by two layered stratified cuboidal and stratified columnar epithelia.

Identify:

( ) CT septa ( ) Mucus secretory units ( ) intercalated ducts
( ) striated ducts ( ) Interlobar ducts ( ) CT capsule (if present)
( )Serous demilune ( ) Lobule

Submandibular Gland

Slide #36 Submandibular, H&E x175 and H&E x725

Connective Tissue

The CT septa are best observed at low magnification. The interlobular septa are thin, and the matrix is barely visible. As in the previous slide it is readily evident around the larger interlobular ducts.

Glandular Epithelium

This is a **branched tubuloacinar gland** comprised of mostly serous and some mucus cells. Staining and nuclear characteristics should be used to differentiate the serous from the mucus secretory units. The mucus secretory glands are mostly of the tubular variety. **Serous demilunes** can be found along the basal margins of the mucus epithelial cells.

Duct Epithelium

Simple and stratified intralobular ducts can be found within the glandular clusters. Interlobular ducts are found in the connective tissue septa. Duct cells are relatively eosinophilic and smaller ducts may exhibit basal striations. In striated interlobular ducts the basal cells possess larger, more euchromatic nuclei than are present in the very eosinophilic apical cell layer.

Identify:

( ) Mucus secretory unit ( ) Serous demilune ( ) CT septa ( ) Lobule
( ) intercalated duct (low cuboidal epithelium with wide lumen)
( ) Intralobular Duct ( ) Interlobular duct

Parotid Gland

Slide #37 Parotid, H&E x160 and H&E x 640

Connective tissue

The CT septa are best observed at low magnification. The interlobular septa are thin and lightly stained, though it is readily evident around the larger interlobular ducts.

Glandular Epithelium

The parotid gland is a **branched acinar gland** comprised almost exclusively of serous secretory units. The densely staining grape-like acini are enmeshed within a non-staining adipose (fat cell) rich mesenchyme. The basophilic protein secreting acinar cells usually contain **zymogen granules.**

Duct Epithelium

In the lobules, most of the obvious ducts range from **simple cuboidal intercalated ducts** (about the diameter of a secretory unit or smaller but with a wide lumen) to **simple columnar intralobular striated ducts.** The larger **interlobar ducts** are usually found within dense interlobar CT matrices.

Identify:

( ) CT capsule	**( ) CT septa**	**( ) Serous acini**	**( ) Intercalated duct**
( ) Intralobular duct	**( ) Unilocular adipose**	**( ) Lobule**	